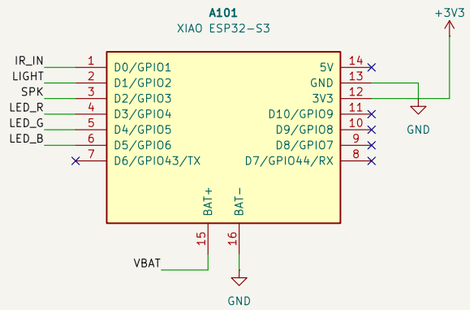
CamoBots — HIDER BODY

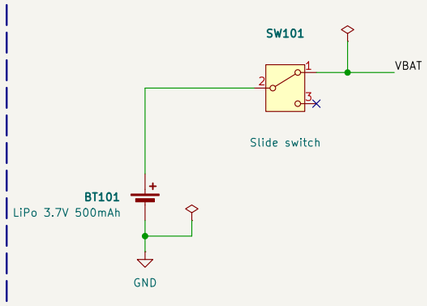
IR receiver catches the blaster's shot > light sensor catches cheating cover > speaker whistles > RGB LED shows the state

HOST (XIAO ESP32-S3)



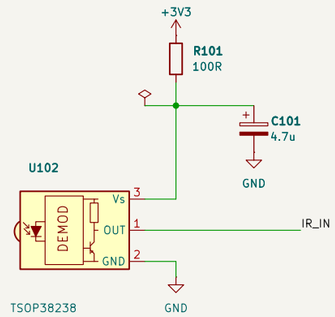
D0–D5 are ADC1 pins: safe to read with Wi-Fi on.
D2 is a boot strapping pin: the 10k pull-down keeps it low.
If boot misbehaves, move the speaker to D8 (GPIO7).

BATTERY + SWITCH



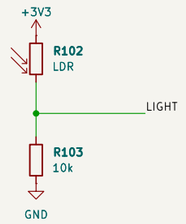
Switch sits in the battery + lead.
VBAT = battery after the switch,
to the BAT+ pad under the XIAO.
Charge through the XIAO USB-C port.
The 5V pin is dead on battery, so
loads that need more than 3.3 V use VBAT.

IR RECEIVER



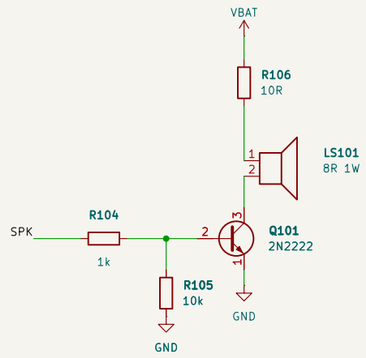
OUT idles HIGH and pulses LOW on a 38 kHz burst.
100R + 4.7u filter its supply.
Dome facing you, legs down: OUT GND VS.
Never paint over the dome.

LIGHT SENSOR



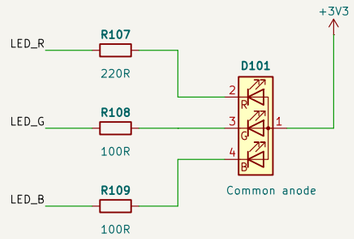
More light = lower LDR resistance
= higher reading on D1.
Room light ~900, covered ~100.
Below 40% of the start level
counts as covered.

SPEAKER DRIVER



D2 HIGH switches the transistor on and the speaker sounds.
1k limits base current; 10k holds it off while the XIAO boots.
10R keeps speaker current safe. Speaker + goes to VBAT, not 5V.
2N2222 / S8050: flat face to you, legs down = E B C (check yours).

RGB STATUS LED



Longest leg is the common anode: it goes to 3V3.
A pin driven LOW turns its colour ON.
Red has a higher resistor because it is brighter.
Teal = waiting, blue = hiding, red = caught,
rainbow = survived the round.

PARTS (ONE BODY)

A101	Seeed XIAO ESP32-S3 + 2.4 GHz antenna
U102	TSOP38238 38 kHz IR receiver
R102	LDR 5 mm (light sensor)
Q101	2N2222 or S8050 NPN transistor
LS101	Small speaker, 8 ohm 1 W (TR-WS-2014B)
D101	5 mm RGB LED, common anode
BT101	LiPo 3.7 V 400–500 mAh, protected
SW101	Slide switch
C101	4.7 uF electrolytic
R101	100R
R103 R105	10k
R104	1k
R106	10R
R107	220R
R108 R109	100R

Wire and test each block on USB first, fit the battery and switch last.

Drawn by Abhinav Krishna — CamoBots
Sheet 1 of 2 | designators 1xx
TSOP38238 + LDR + 2N2222 speaker driver + common-anode RGB LED
Single-cell LiPo, everything runs from the XIAO 3V3 pin except the speaker
CamoBots | Designed by Abhinav Krishna

Sheet: /
File: body.kicad_sch

Title: Hider Body

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1